

Jordan University of Science & Technology**Faculty of Science and Arts****Department of Mathematics and Statistics****Year:** 2024/2025**Semester:** Fall 2024**Course Information**

Course Title	Mathematical Applications of Biological Sciences
Course Number	MATH 103
Prerequisites	
Course Website	https://elearn.just.edu.jo/
Instructors	Dr. Nazek Obeidat
Office Location	PH2 3 rd floor
Office Phone	7201000 Ext. 23452
Office Hours	Monday & Wednesday 11:30 – 1:00, Mon., or by appointment
E-mail	Class20@just.edu.jo

Textbook

Title	The material to be covered appears in two or more of the list of references.
References	1. Any Calculus textbook. 2. Frank R. Giordano, Maurice D. Weir, William P. Fox, A First Course in Mathematical Modeling, 4th Edition. 3. Earl Rainville, Phillip Bedient and Richard Bediant, Elementary Differential Equations, 8th edition, Macmilan. https://www.abebooks.com/book-search/title/elementary-differential-equations/author/phillip-e-rainville-earl-d/ 4. Dennise Zill, A First Course in Differential Equations with Modeling Applications, 9th edition.. https://www.academia.edu/33171896/_Dennis_G_Zill_A_First_Course_in_Differential_Equations 5. Finite Mathematics, 6th ed., Daniel Maki, Maynard, Thompson, Stephen McKinley, 2017 https://ebin.pub/qdownload/finite-mathematics-6th-edition-6nbsped-1259819760-9781259819766.html 6. Applied Finite Mathematics Third Edition, 2016 Revised and Updated by Roberta Bloom. https://www.deanza.edu/faculty/bloomroberta/documents/AppliedFiniteMath-3ed-Current.pdf

Assessment Policy

Assessment Type	Expected Due Date	Weight
First Exam	According to the university calendar	30%
Second Exam	According to the university calendar	30%
Final Exam	According to the university calendar	40%

Class Schedule & Room

Lecture Time: Section 1: Monday & Wednesday 10:00 – 11:30 Room: NG 41

Course Catalog

3 Credit Hours. This course is design for the Biological Sciences students. First, we give the students an overview of the structure and applications of functions, differentiation, integration, then, first order differential equations, Modelling with ordinary differential equations: population growth (logistic and harvesting), more applications, Modelling change with difference equations, Model Fitting: graphically, least squares, linear programming, the diet problem.

Teaching & Learning Methods

- Class lectures and exams are designed to achieve the course objectives.
- You should read the assigned chapters before class and participate in class and do whatever it takes for you to grasp this material. Ask questions. Ask lots of questions.
- You are responsible for all material covered in the class.
- Please communicate any concerns or issues as soon as practical either in class, or by Email.

Goals/Objectives

The main goal of this course is to give the student a brief introduction to the basic mathematical concepts and some of their properties, will be useful to develop knowledge of the mathematical language and to study the applications to various practical situations. We will constraint on the biological real-life problems.

an overview of various concepts of Mathematics, the structure and applications of functions, differentiation, integration, ordinary differential equations, Laplace transformation and its for ODEs, and applications of linear differential equations to Pharmacology.

Mapping of Course Learning Outcomes to Student's Learning Outcomes At the end of the course the student be able to	Outcome Weight	Link to SO's	Linked with NQF Descriptors	Assessment method
1. understand the basic (fundamental) mathematical concepts.	40%	30 SO1, 10 SO2	1L7K1, 1L7S1, 1L7S2, 1L7S3, 1L7C1	Exams and Quizzes
2. gain proficiency in correct mathematical reasoning and to apply the mathematical tools to practical situations.	30%	10 SO1, 5 SO2, 15 SO3	1L7K1, 1L7S1, 1L7S2, 1L7S3, 1L7C1, 1L7C2, 1L7C4	Exams and Quizzes
3. understand the complex nature of developing biological models: what they are, how they are constructed, and how they are interpreted and, as a result, should have an appreciation for the power of mathematics.	30%	10 SO1, 10 SO2, 10 SO3	1L7K1, 1L7S1, 1L7S2, 1L7S3, 1L7C1	Exams and Quizzes

Relationship to Student Learning Outcomes (Out of 100%)

SLO1	SLO 2	SLO 3	SLO 4	SLO 5	SLO 6
50	25	25			

Course Content

Chapter #	Subject	Week #
1	Review of functions, Logarithmic and exponential functions, trigonometric functions.	1 + 2
2	Review of integration: Indefinite Integrals, Definite Integrals, Techniques of Integrations; Substitutions, integration by parts, integration of rational functions.	3
3	First order differential equations; existence and Uniqueness Theorem, separable equations, linear equations, exact equations and Special Integrating Factors.	4+5
4	Applications of linear differential equations to Biology such as the growth law	6+7

	(Logistic model with and without harvesting) and the cooling law.	
5	Modelling Fitting: graphically and least squares.	8
6	Basic Statistics	9
7	Basic Probability	10
8	Introduction to matrices and Applications	11+12
9	Introduction to Linear Programming, examples: Disease and Diet problems	13+14
10	Review	15

Additional Notes	
Exams	- The format for the exams is generally (but NOT always) as follows: Problem formulation, computation, analytic solutions, and analysis. - Grades will not be given out via e-mail. - The final exam covers all the material in the course.
Quizzes	
Projects and Reports	
Makeup Exams	- Let the instructor know about your makeup exam before 3 days prior to the scheduled exam time. - Makeup exam should not be given unless there is a valid excuse.
Drop Date	According to the university calendar.
Cheating	- Cheating or copying from neighbor on exam, or quiz is an illegal and unethical activity. - Standard JUST policy will be applied.
Attendance	- Excellent attendance is expected. - JUST policy requires the faculty member to assign ZERO grade (35) if a student misses 20% of the classes that are not excused. - Sign-in sheets will be circulated. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
Workload	Average work-load student should expect to spend is 4 hours/week.
Graded Exams	Instructor should return in term exam, quiz, HW, and project papers graded to students during the week after the due date.
Participation	Participation in, and contribution to class discussions will affect your final grade positively. Raise your hand if you have any question.
Finally	Smoking is prohibited in all in-door places.

Dr. Nazek. Obeidat

Last modified: October 21, 2024.